

Chuyot Liewlam 6701134

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

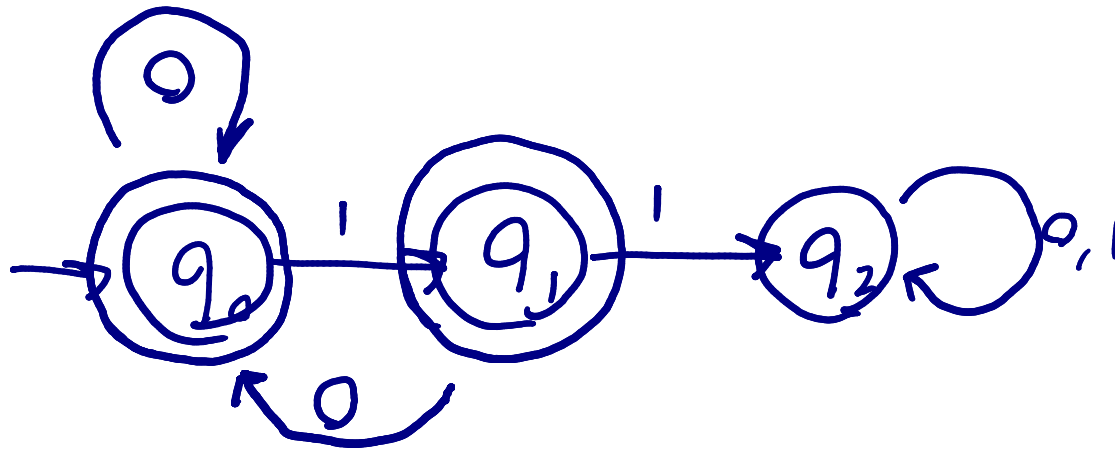
Theory of Computation

Exercise 2: (Deterministic Finite Automata - DFA)

1. Draw DFA for $L1$

$L1 = \{w \in \{0,1\}^* : w \text{ has no substring } 11\}$

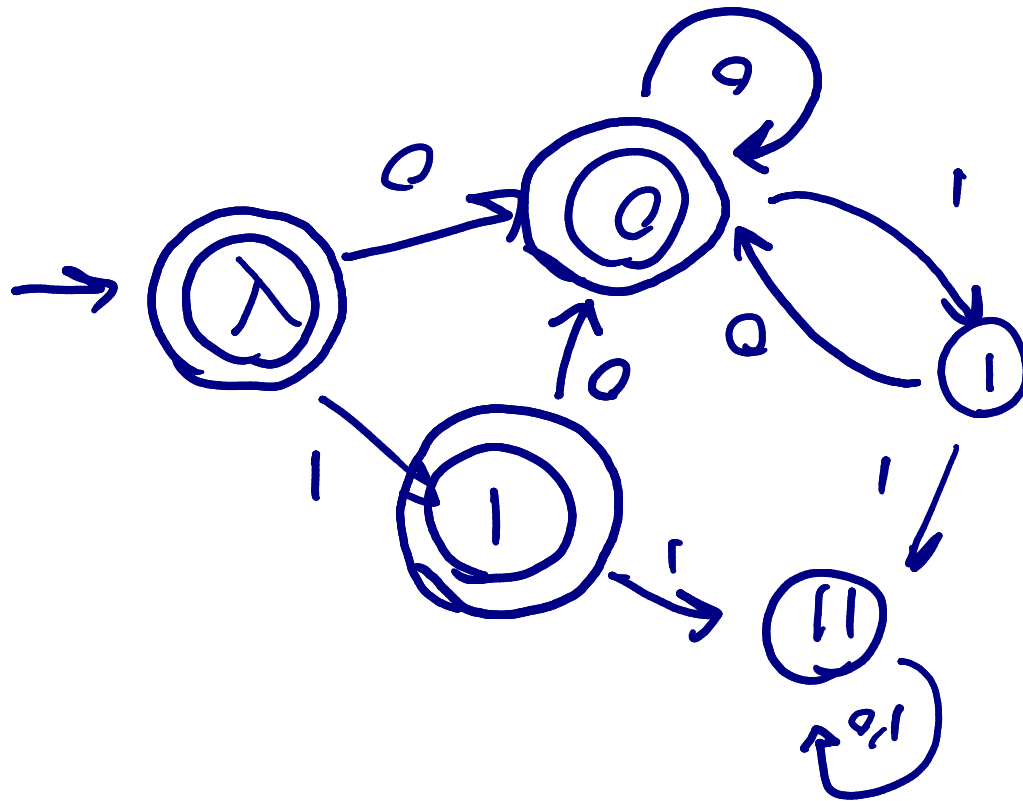
$\Sigma^* \Rightarrow \{\lambda, 0, 1, 00, 01, 10, 000, 001, \dots\}$



$\Sigma = \{0, 1\}$

2. Draw DFA for L2

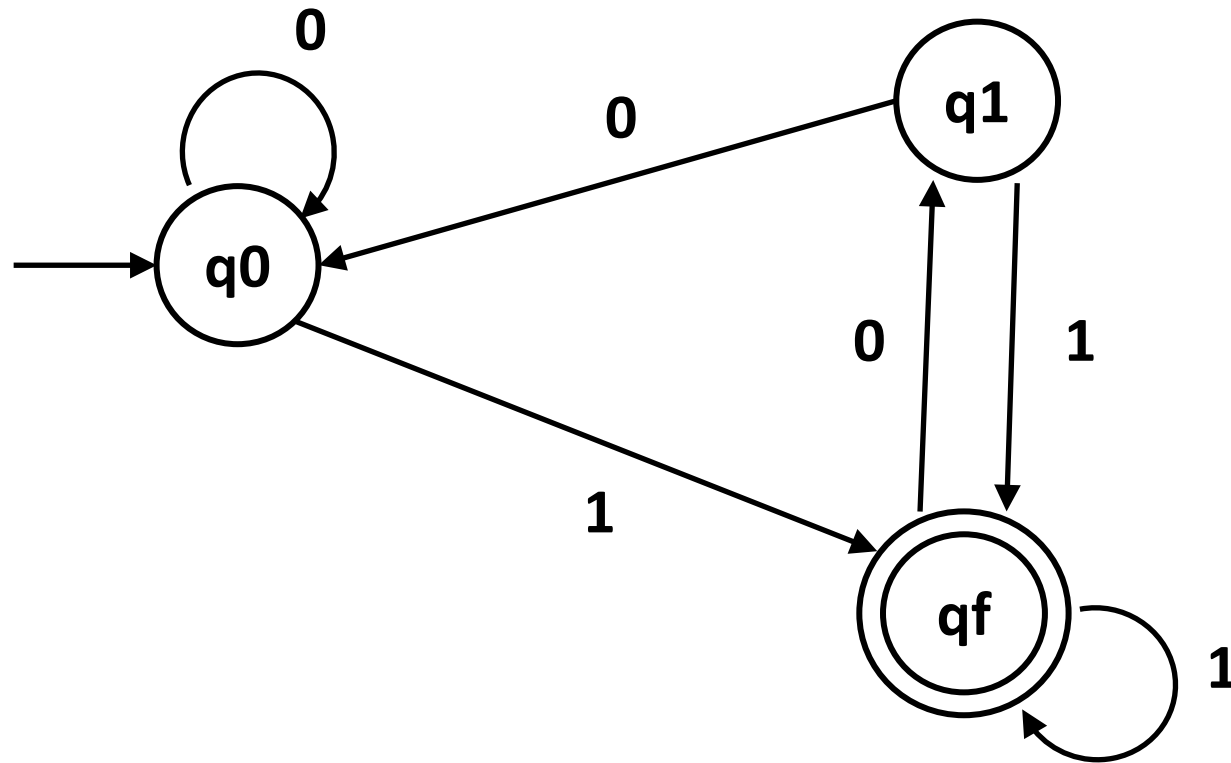
$$L2 = \{0, 10\}^* \cup \{1\}$$



3. Find the language of DFA M.

$\{01, 001, 011, 0001, 01101, 0111, \dots\}$

M:



Suffix 1

* 4. Draw DFA for L3

$$L3 = \{ a^m b^n : m + n = 5; m \text{ and } n \geq 0 \}$$

